Why does Multilingual Training improves Theorem Proving?

A Small-Scale Controlled Study

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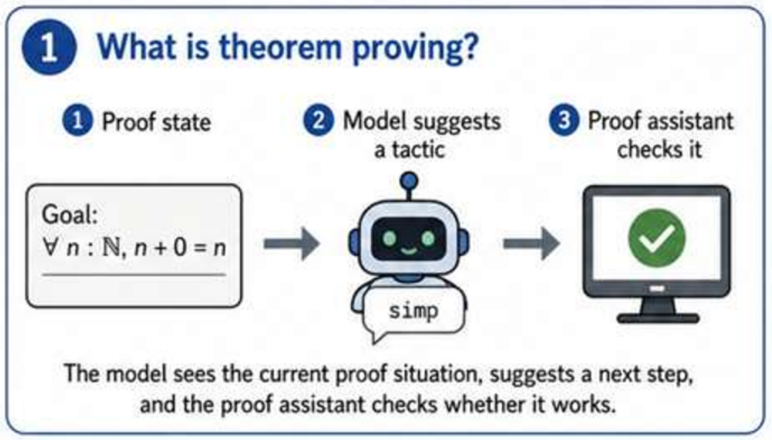
Neural Theorem Proving | Proof Search | Multilingual Training



ProofWala result:

Multilingual training improves it!

Research Question

Why does multilingual training improve theorem proving performance?

H1: Cross-language Transfer

Models learn shared semantic patterns across proof systems.

Semantics

VS

H2: Regularization

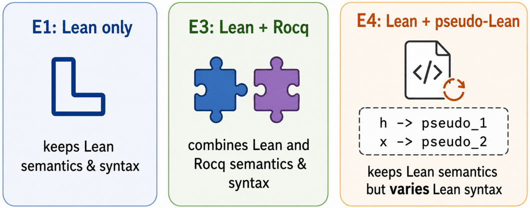
More varied syntactic training data improves robustness.

Syntax

Our Training Experiment

Three Controlled Training Setups

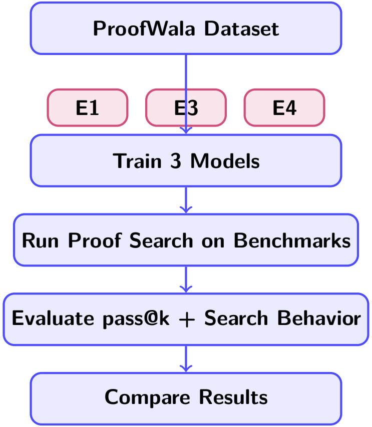
Model	Training Data	Purpose
E1	Lean only	Baseline
E3	Lean + Rocq	Real multilingual
E4	Lean + pseudo-Lean variant	Regularization control



Methodology

Shared Setup

- Same architecture: CodeT5-small
- Same tokenizer
- Same number training steps
- Same proof-search algorithm
- Same evaluation benchmarks



Benchmarks:

Easy_Lean
Set of easy-to-prove Lean theorems whose proofs share similar semantic patterns

Hard_Lean
Set of harder-to-prove Lean theorems with more diverse semantic proof patterns

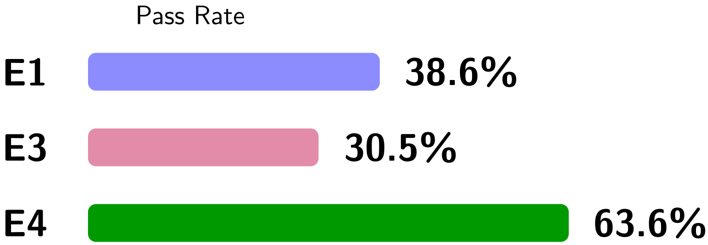
Key Experimental Design

We changed only the training data.

This isolates the effect of multilingual vs. pseudo-multilingual training and strengthens causal interpretation.

Main Results

Easy_Lean Results



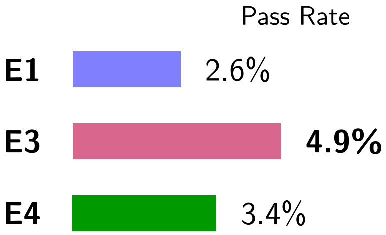
E4 ≫ E1 > E3

Interpretation

Only-syntax variation helped more than syntax variation & combined semantics on problems whose solutions have similar combinations of steps!

Main Results

Hard_Lean Results



E3>E4>E1

Interpretation

Syntax variation & combined semantics may help more than only-syntax variation on problems that require solutions with fundamentally different ideas.

Main Interpretation

Our benchmarks reveal different mechanisms

Easy_Lean

E4 wins

Pseudo-multilingual

$\Downarrow$

H2 is supported **when** solutions have similar combinations of steps

Hard_Lean

E3 wins

Real multilingual

$\Downarrow$

H1 is supported **when** solutions require fundamentally different ideas

Main Conclusion

H1 and H2 hold under different circumstances.

Key Findings

Main Experimental Findings

- Multilingual gains are not driven by a single mechanism
- Pseudo-multilingual training (i.e., only-syntax variation) can improve performance on theorems whose proofs have similar combinations of tactics.
- Real multilingual training (i.e., syntax variation and combined semantics) may improve performance on theorems whose proofs require newer combinations of tactics.

Limitations & Future Work

Limitations

- Small-scale pilot study
- Limited compute budget
- Small Hard_Lean evaluation
- Pseudo-Lean transformations are imperfect

Future Work

- Larger multilingual datasets
- Deeper proof-search analysis
- Stronger theorem proving models

Main Perspective

This work is an initial step toward understanding **why multilingual training helps theorem proving**.

Thank you!